CONFIDENTIAL

UNIVERSITI TEKNOLOGI MALAYSIA

FINAL EXAMINATION SEMESTER 2, 2025/2026

SUBJECT CODE	**:**	**SCSE2233**
SUBJECT NAME	**:**	**SOFTWARE DESIGN & ARCHITECTURE**
DATE / DAY	**:**	
DURATION	**:**	**3 HOURS**
VENUE	**:**	

INSTRUCTIONS

Answer all questions in the Answer Booklet. All diagrams must be drawn using the UML standard notations by hand. Note that any form of exam misconduct caught during the exam will cause students to get a zero in this assessment. This question booklet needs to be returned with the Answer Booklet and Answer Sheets (if any) by the end of the examination. You are only allowed to open this booklet with the permission of the examination invigilators.

(Please Write Your Lecturer Name And Section In Your Answer Booklet)

Name	
I/C No.	
Year / Course	
Section	
Lecturer Name	

This question paper consists of 6 printed pages excluding this page.

Mock examination paper generated for personal revision practice — not an official UTM examination paper.

SECTION A - STRUCTURED QUESTIONS

INSTRUCTIONS: Answer ALL of the following questions.

Question 1 **(16 marks)**

a) Assume that you have **FIVE (5)** team members to complete the project based on the given case study in Figure 1. Draw a WBS (Work Breakdown Structure) together with a Linear Responsibility Chart (LRC) that distributes the tasks among the five of you using the standard responsibility codes (1 = Primary responsibility, 2 = Support role, 3 = Review role). The proposed WBS must satisfy the **THREE (3)** WBS evaluation criteria — scope accuracy, completeness, and level of detail — to ensure the WBS includes the complete set of tasks necessary to complete the design project.

> Universiti Teknologi Malaysia wants to develop CampusRide, a smart shuttle-tracking platform for its campus shuttle fleet. The platform must show students the live GPS position of every shuttle, predict arrival times at each stop, display seat availability, and let students plan a route between any two points on campus. Drivers use a separate console to start and end trips, while the transport office uses an analytics dashboard to monitor fleet utilisation and adjust schedules. The platform must support role-based secure access for students, drivers, and transport-office administrators.

Figure 1: Case Study CampusRide Smart Shuttle-Tracking Platform

(6 Marks)

b) You are managing the CampusRide software design project composed of **TWELVE (12)** tasks. Each task is estimated to cost RM 1,500, and the total Budget at Completion (BAC) is RM 18,000. The project is planned to take **TWELVE (12)** months, with each task planned to be completed in one month. At the end of the current 7th month, only **FOUR (4)** tasks have been fully completed, at an actual cost of RM 7,500. Using the Earned Value Management (EVM) equations provided in Appendix A, calculate EV, PV, CV, SV, CPI, SPI, ETC, and EAC, and interpret the meaning of your CPI and SPI values for upper management in ONE (1) sentence each.

(6 Marks)

c) During the 7th-month progress review, your project manager instructs you to report the project to the client as “on schedule and on budget”, and to record the three partially started tasks as fully completed so that the reported figures look healthier.

i) Based on the IEEE-CS/ACM Software Engineering Code of Ethics (SECE), identify **ONE (1)** principle that this instruction violates, and explain how it is violated in this situation.

(2 Marks)

ii) There are FOUR (4) keys in leadership skills. Choose **ONE (1)** of them and explain how you would apply it to resolve this situation professionally.

(2 Marks)

Question 2 **(12 marks)**

a) List **TWO (2)** key quality attributes that are commonly evaluated in software design construction to ensure the quality of the design.

(2 Marks)

UTMFood is a campus food-delivery service. The delivery charge and the reward perk for each order are decided as follows. Customers are classified by membership tier: BASIC, GOLD, or PLATINUM. Orders are classified by order value: below RM 50, or RM 50 and above. A BASIC member pays a RM 5 delivery fee for an order below RM 50 and receives no perk; for an order of RM 50 and above, the fee is RM 3 and a free drink is added. A GOLD member pays RM 3 with a free drink for an order below RM 50, and RM 1 with a free dessert for an order of RM 50 and above. A PLATINUM member pays RM 1 with a free dessert for an order below RM 50; for an order of RM 50 and above, delivery is free (RM 0) and the order receives a free dessert together with priority delivery.

Figure 2: UTMFood Delivery Charge and Perk Policy

b) Name the type of decision table that is best suited to represent the policy in Figure 2 and justify your choice. Then, using the appropriate formula, compute the number of distinct elementary policies that the table must contain.

(2 Marks)

c) Draw the complete Extended Entry Decision Table (EEDT) for the policy described in Figure 2.

(5 Marks)

d) Write a source-code excerpt (in any programming language taught in this course) that implements the decision logic of your EEDT in (c).

(3 Marks)

Question 3 **(12 marks)**

a) PDL (Programming Design Language) popularity stems from the use of natural language, as opposed to computer languages or graphical techniques, to define the required behaviour of functions, providing a “comments first” approach to constructing code. Based on the SmartLib self-checkout function described in Figure 3, write a PDL that results in a detailed function design that is easy to create, easy to review, and easy to translate to code.

The SmartLib self-checkout function begins by reading the student's matric card. If the card is invalid, the function displays an error message and ends the session. If the card is valid, the system retrieves the member record and checks the member's outstanding fines. If the fines exceed RM 10, the loan session is rejected and the member is directed to the service counter. Otherwise, for each book placed on the scanner pad, the system reads the book's RFID tag and verifies that the book is not reserved by another member. A reserved book is set aside and

reported on the screen; a non-reserved book is registered as a loan with the due date set to 14 days from today. After all books are processed, the function prints a receipt listing every successful loan with its due date, and returns a value indicating the success or failure of the session.

Figure 3: SmartLib Self-Checkout Function Description

(4 Marks)

b) Based on the sample code in Figure 4, convert the code into a program flow graph. Label every node and edge clearly.

```
1  import java.util.Scanner;
2  public class ChargePoint {
3    public static void main(String[] args) {
4      Scanner scanner = new Scanner(System.in);
5      int kWh;
6      do {
7        System.out.print("Enter energy used (1-100 kWh): ");
8        kWh = scanner.nextInt();
9      } while (kWh < 1 || kWh > 100);
10     System.out.print("Enter user type (1=Student, 2=Staff, 3=Public): ");
11     int type = scanner.nextInt();
12     double rate;
13     switch (type) {
14       case 1:  rate = 0.20; break;
15       case 2:  rate = 0.35; break;
16       default: rate = 0.50;
17     }
18     double cost = kWh * rate;
19     if (kWh > 60 && type != 3) {
20       cost = cost * 0.90;
21     }
22     for (int i = 0; i < 2; i++) {
23       System.out.println("Receipt copy " + (i + 1));
24     }
25     System.out.println("Total charge: RM " + cost);
26     scanner.close();
27   }
28 }
```

Figure 4: Sample Code of EV Charging Fee Calculator

(4 Marks)

c) Determine the cyclomatic complexity of the code in Figure 4 using **TWO (2)** different approaches that you have learned, and show that both approaches produce the same value.

(4 Marks)

SECTION B - PROBLEM SOLVING

INSTRUCTIONS: Answer ALL the following questions.

Question 4 **(30 marks)**

Project 1: TripCraft Travel-Package Configurator

TripCraft assembles customised travel packages. Every package is constructed through the same fixed sequence of steps — reserve transport, reserve accommodation, add activities, and attach travel documents — but the content of each step differs between package tiers. The same construction sequence must be able to produce different final representations, such as a Budget package and a Luxury package, and the construction process must be controlled step by step so that optional steps can be skipped. The completed package object is retrieved only at the end of the construction process. New package tiers are planned for the future.

Project 2: MediTrack Ward Equipment Dashboard

MediTrack supplies hospital wards with equipment-monitoring software. Two product lines exist: an ICU line and a General-Ward line. Each line provides a family of related components — a VitalsMonitor, an InfusionPump, and an AlarmPanel — and components from the same line must always be used together and remain consistent (an ICU VitalsMonitor must never be paired with a General-Ward AlarmPanel). The hospital must be able to switch an entire ward from one product line to the other without modifying the client code, and new product lines (for example, a Pediatric line) will be introduced in the future.

a) Analyze the scenario of both projects and identify a suitable Creational detail design pattern for each project above. Justify both chosen Creational detail design patterns.

(6 Marks)

b) Choose **ONE (1)** project and design the class diagram for the Creational detail design pattern. The class diagram must reflect your Creational detail design pattern answered in (a).

(10 Marks)

c) Elaborate your diagram in (b); describe the components of the diagram.

(4 Marks)

d) Also based on the class diagram answered in (b), evaluate and write the code representing the class diagram.

(10 Marks)

Question 5 **(30 marks)**

Project 1: FarmSense Greenhouse Monitoring System

FarmSense controls a smart greenhouse. A climate controller continuously monitors temperature and humidity readings. Whenever a reading crosses a configured threshold, ALL registered parties must be updated automatically and immediately: the dashboard display, an SMS alert service, an email service, and an automatic roof-vent controller. Farm managers must be able to add or remove notified parties dynamically at run time, and new kinds of parties (for example, a mobile push-notification service) must be supported in the future without modifying the climate controller.

Project 2: SwiftParcel Customs-Clearance Integration

SwiftParcel's dispatch system processes parcels through an internal interface that declares the operations trackParcel(), registerParcel(), and scanParcel(), and this interface is used throughout the dispatch system's code. The company has now licensed a third-party customs-clearance module that is delivered only as a compiled library (no source code available), exposing the operations trackCustomsQR(), registerCustomsQR(), and scanCustomsQR(). The dispatch system must be able to use the customs-clearance module anywhere the internal parcel interface is expected. Neither the dispatch system's existing client code nor the third-party library may be modified.

a) Analyze the scenario of both projects and identify a suitable detail design pattern for each project above. For EACH chosen pattern, state its pattern category (creational, structural, or behavioral) and justify your choice.

(6 Marks)

b) Choose **ONE (1)** project and design the class diagram for the detail design pattern. The class diagram must reflect your detail design pattern answered in (a).

(10 Marks)

c) Elaborate your diagram in (b); describe the components of the diagram.

(4 Marks)

d) Also based on the class diagram answered in (b), evaluate and write the code representing the class diagram.

(10 Marks)

- END OF QUESTIONS -

APPENDIX A

Table 1: Earned Value Management (EVM) Definition of Parameters

Parameter	Description
EV	Also called budgeted cost of work performed (BCWP), it is the dollar amount of the work currently completed
AC	Also called actual cost of work performed (ACWP), it represents the amount of money that was spent in the work currently completed
PV	Also called budgeted cost of work scheduled (BCWS), it represents the value of the work that was expected to be completed at that point in time of the schedule
CV	Describes the difference between the value of work completed and the actual cost spent to complete it
SV	Describes the difference between the value of work currently completed and the cost of the work that was expected to be completed
CPI	Describes the current cost efficiency of the work (CPI < 1 means over budget; CPI > 1 means under budget; CPI = 1 means right on budget)
SPI	Describes the current schedule efficiency of the work (SPI < 1 means behind schedule; SPI > 1 means ahead of schedule; SPI = 1 means right on schedule)
BAC	Total budget for the project
ETC	Represents the estimated cost remaining to complete the project
EAC	Represents the estimated total cost of the entire project

Table 2: Earned Value Management (EVM) Equations

Parameter	Equation
EV	EV = (% work completed) × planned cost for the work
CV	CV = EV − AC
SV	SV = EV − PV
CPI	CPI = EV / AC
SPI	SPI = EV / PV
ETC	ETC = (BAC − EV) / CPI
EAC	EAC = ETC + AC